Coset residues and chamber orbits

Human Compression project

Abstract

We give a complete account of the valid residue arguments and the necessary corrections in Leemans–Tranchida's paper on residual connectedness. The rank-three result and a criterion using the specified group's action are proved. The published general criterion, which uses chirality under the full automorphism group, has a previously verified counterexample. Two smaller examples dispose of auxiliary assertions whose wording omits firmness or a rank restriction. The counterexamples, their scope and the corrected theorems are kept distinct. This manuscript assembles prior project proofs and the accepted corrections; it makes no additional conjecture or measurement claim.

1 Source and conventions

The source is the final journal edition of Leemans–Tranchida [1], pages 491–499. Its principal Theorem 1.1 asserts residual connectedness for the standard coset geometry of a C^+-group if that geometry is chiral. Chirality is defined using the *full* type-preserving automorphism group. That group need not be the specified C^+-group. The accepted five-page manuscript [4] gives an explicit counterexample with specified group $\mathrm{PSL}_2(13)$ and full group $\mathrm{PSL}_2(13) \times C_2$. It is reproduced in the complete volume with its proof unchanged.

We retain the source's definition of a geometry: every flag extends to a chamber. Firmness means that every panel has at least two completions; thinness means exactly two. Residual connectedness asks that every residue of rank at least two, including the whole geometry, have a connected incidence graph. None of these extra conditions is implicit in the word geometry. In particular, a rank-one residue with two objects has a disconnected incidence graph.

Let I be a finite type set. We use left cosets gH_i and incidence by nonempty intersection, retaining the type as part of an object. The source uses right cosets. Inversion sends $H_i g$ to $g^{-1}H_i$ and is an isomorphism of the two incidence systems. Thus this convention changes no flags, residues or transitivity assertions. Write $H_J = \bigcap_{j \in J} H_j$, with $H_\varnothing = G$. All general proofs below permit infinite groups.

The coset construction, the C^+ intersection condition and the canonical neighbouring chambers are prior results of the source and its cited predecessors. Sections 2–5 rework the existing project arguments [2, 3]. The chamber-component explanation in Section 6 is also retained from [3]. The principal counterexample [4], the rank boundary [5], and the seven-object example [6] have distinct reviewed proofs. Classical orbit arguments are not claimed as new. No new Lean proof is asserted by this assembly.

2 Flags and the internal residue map

The base objects H_i form a chamber. Each object is a translate of one of them. If gH_i and hH_j are incident, choose an element x of their intersection. Their pair is then $x\{H_i, H_j\}$. This proves the object and pair transitivity in the cited Tits construction, and also proves that every flag of rank at most two extends to a chamber.

Fix a proper subset $J \subset I$, and let $F_J = \{H_j : j \in J\}$. For $i \notin J$, put

$$S_{J,i} = \bigcap_{j \in J} H_j H_i.$$

For $J = \varnothing$ the intersection is G. The type-i objects of the residue of F_J are exactly the cosets gH_i with $g \in S_{J,i}$.

Lemma 2.1. *The maps*

$$h(H_J \cap H_i) \longmapsto hH_i \qquad (h \in H_J, i \notin J) \tag{1}$$

define a homomorphism from the internal coset incidence system $\Gamma(H_J; (H_J \cap H_i)_{i \notin J})$ *into the residue of* F_J*. It is injective on objects. It is surjective on objects if and only if* $S_{J,i} = H_J H_i$ *for every* $i \notin J$*.*

Proof. For $h, k \in H_J$, equality of hH_i and kH_i is equivalent to $k^{-1}h \in H_i$, and hence to $k^{-1}h \in H_J \cap H_i$. This proves both well-definedness and injectivity. A common element of two internal cosets is a common element of the corresponding external cosets, proving preservation of incidence. The image at type i consists of the cosets represented by $H_J H_i$. The full residue at that type is represented by $S_{J,i}$, and both sets are unions of left cosets of H_i. Equality is therefore exactly the surjectivity condition. □

A bijective incidence homomorphism need not be an isomorphism: its inverse may fail to preserve incidence. This distinction in the source's Remark 4.4 is essential. For a singleton $J = \{j\}$ the displayed surjectivity condition is automatic, but no assertion about the inverse follows from this alone.

Lemma 2.2. *Suppose the coset system is a geometry. The action of* G *is transitive on flags of type* $J \cup \{i\}$ *if and only if it is transitive on flags of type* J *and* $S_{J,i} = H_J H_i$*.*

Proof. Transitivity on the larger type implies transitivity on the smaller type, because every smaller flag has an extension in a geometry. For $g \in S_{J,i}$, the flag $F_J \cup \{gH_i\}$ is of the larger type. An element carrying the corresponding base flag to it fixes every member of F_J, so lies in H_J. Hence $gH_i = hH_i$ for some $h \in H_J$ and $g \in H_J H_i$. The reverse inclusion always holds.

Conversely, normalize the J-part of any flag of the larger type to F_J. Its last object is gH_i with $g \in S_{J,i}$. Write $g = hu$ with $h \in H_J$ and $u \in H_i$. Multiplication by h^{-1} fixes F_J and takes the last object to H_i. Thus every such flag is a translate of the base flag. □

This is the source's Lemma 4.5. If the maps (1) are surjective for every J, Lemma 2.2, by induction on the size of a flag, gives flag transitivity. In that case (1) is an isomorphism: any incident pair in a residue, together with F_J, is a flag and can be normalized by an element of H_J. That element is a common representative in the internal model. This proves the final implication in the cited source Lemma 4.3 for the maps used here.

Surjectivity for just one J does *not* supply transitivity on J. This is the missing condition in Proposition 4.6. The retained counterexample is $G = C_2^2$, with its three order-two subgroups H_0, H_1, H_2 and $H_3 = G$. Its geometry is the cone over the complete tripartite graph $K_{2,2,2}$, with seven objects and eight chambers. For $J = \{0, 1, 2\}$ the base residue and its internal coset model each have one object, so (1) is surjective. But a group of order four cannot be transitive on the eight J-flags. This is a counterexample to the stated equivalence, not to a C^+ assertion.

3 Connectivity and the specified action

Lemma 3.1. *Let K_i be subgroups of a group K and let the index set have at least two elements. The incidence graph of their coset system is connected if and only if $K = \langle K_i : i \in I \rangle$.*

Proof. The base cosets form a clique. Each K_i fixes one of its vertices, and therefore preserves the connected component of that clique. If the subgroups generate K, every translate of every base vertex lies in this component, proving connectedness. Conversely, a path starting at a base vertex can be followed by successively choosing elements in the intersections of its consecutive cosets. Each step changes the representative by multiplication by an element of one of the K_i. Every vertex reached therefore has a representative in $\langle K_i : i \in I \rangle$. If all vertices are reached, each coset of any one K_i has such a representative, so their union gives $K = \langle K_i : i \in I \rangle$. □

Proposition 3.2. *Suppose G is transitive on flags of every proper type in a coset geometry. If $|I \setminus J| \geq 2$ and*

$$H_J = \langle H_J \cap H_i : i \in I \setminus J \rangle, \tag{2}$$

then every residue of a flag of type J is connected.

Proof. Normalize the flag to F_J. For every $i \notin J$, the set $J \cup \{i\}$ is still proper, so Lemma 2.2 and Lemma 2.1 make the internal model bijective on objects. Equation (2) and Lemma 3.1 make that internal graph connected. Its image is a connected spanning subgraph of the actual residue. Any extra incidence edges preserve connectedness. □

This is the sufficient residue statement used in the source, with its action and rank hypotheses explicit. In rank one, transitivity on the enlarged type would mean transitivity on chambers, which has not been assumed. The rank-one boundary of its printed Proposition 4.7 is exhibited in [6]. The source's preceding paragraph already excludes these residues from the residual-connectedness question.

4 Canonical chambers of a marked group

From now on let $|I| = r \geq 3$, with a distinguished element 0. Let G be generated by distinct nonidentity elements α_i for $i \neq 0$, and put $\alpha_0 = 1$. Set

$$a_{ij} = \alpha_i^{-1}\alpha_j, \qquad P_S = \langle a_{ij} : i, j \in S \rangle, \qquad H_i = P_{I \setminus \{i\}}.$$

Assume the C^+ intersection condition

$$P_S \cap P_T = P_{S \cap T} \qquad (|S|, |T| \geq 2). \tag{3}$$

For an empty or singleton set S, the group P_S is trivial, so (3) holds also when either index set has size less than two. Repeated intersection therefore gives

$$H_J = P_{I \setminus J}. \tag{4}$$

For $i > 0$, H_i is generated by the α_j with $j \neq i$; for $i = 0$, it is generated by $\alpha_1^{-1}\alpha_j$ for $j \geq 2$. Thus these are precisely the maximal parabolics in Construction 3.2.

Lemma 4.1. *Let $C = \{H_i : i \in I\}$. For $i \in I$ and any $j \neq i$, put*

$$D_i = (C \setminus \{H_i\}) \cup \{a_{ij}H_i\}.$$

This does not depend on j. It is a chamber distinct from C, adjacent to it at type i. Moreover

$$a_{ji}D_i = D_j, \qquad D_i = \alpha_i^{-1}D_0, \qquad D_0 \text{ is } i\text{-adjacent to } \alpha_i C. \tag{5}$$

The two G-orbits GC and GD_0 are disjoint.

Proof. For $j, k \neq i$, the quotient $a_{ij}^{-1} a_{ik} = a_{jk}$ belongs to H_i, proving independence of the choice. The coset differs from H_i because $a_{ij} \neq 1$ and $P_{\{i,j\}} \cap H_i = P_{\{j\}} = 1$. If $\ell \neq i, j$, its representative a_{ij} belongs to H_ℓ. To obtain incidence with H_j, choose $k \neq i, j$; this is where $r \geq 3$ is needed. The element $a_{ik} = a_{ij} a_{jk}$ lies in both $a_{ij} H_i$ and H_j. All unchanged objects meet at the identity, so D_i is a chamber.

For the first identity in (5), at type i the two multipliers cancel; at type j the multiplier produces the altered coset; at every other type it lies in that parabolic. Taking one index to be zero gives the second identity. Translating the adjacency of C, D_i by α_i gives the last assertion in (5). If $gC = D_0$, then g fixes the base panel missing type zero. Its stabilizer is $H_{I \setminus \{0\}} = P_{\{0\}} = 1$ by (4). This would give $D_0 = C$, a contradiction. □

The lemma is the valid rank-at-least-three form of source Lemma 4.8. It describes neighbours of these canonical chambers. It does not normalize an arbitrary chamber to C.

Proposition 4.2. *If the marked coset system is a geometry and its full type-preserving group A has two alternating chamber orbits, then it is thin. For an arbitrary firm geometry with such a group action, A is transitive on flags of every proper type, and residual connectedness can be tested on the residues of a single chamber.*

Proof. In the marked case, C and D_0 are adjacent and therefore lie in the two different A-orbits. Lemma 4.1 supplies a neighbour of every type for both orbit representatives. Transport by A supplies one for every chamber. Every panel has at least two completions. It cannot have three: three chambers completing one panel are pairwise adjacent, and three pairwise adjacent chambers cannot lie alternately in just two orbits. Thus the geometry is thin.

For the general firm case, let F be a flag of proper type and choose a chamber containing it. At a type missing from F, firmness gives an adjacent chamber still containing F. These two chambers lie in the two different orbits. Given another flag of the same type, extend it to a chamber and compare with the first or second chamber as required. The resulting automorphism maps the whole typed flag to the other one. Thus A is transitive on each proper flag type, and their residues are isomorphic. Every proper type occurs in a chosen chamber; this proves the residue test, including all residues of rank at least two. □

This proves the useful firm versions of source Theorem 4.1 and Proposition 4.2. It does not prove their unrestricted wording: the seven-object example in [6] satisfies the source's literal chirality definition and refutes both. Nor does this proposition give proper-flag transitivity for the subgroup G.

5 The valid residual-connectedness criterion

Theorem 5.1. *For the nondegenerate C^+ marking above, suppose its coset system is a geometry and G is transitive on flags of every proper type. Then the geometry is firm and residually connected.*

Proof. Each panel can be normalized to a base panel. Its two canonical completions from Lemma 4.1 give firmness. Let F_J be a base flag, put $S = I \setminus J$, and assume $|S| \geq 2$. As in Proposition 3.2, the internal residue model is bijective on objects and preserves incidence.

If $|S| \geq 3$, equation (4) identifies its ambient group with P_S and its parabolics with $P_{S \setminus \{i\}}$ for $i \in S$. Every generator a_{uv} of P_S belongs to one of those parabolics: choose $i \in S \setminus \{u, v\}$. They therefore generate P_S, and Proposition 3.2 proves connectedness.

If $S = \{u, v\}$, write $a = a_{uv}$, so $H_J = \langle a \rangle$. Its two internal parabolics are trivial. The residue has two copies of this cyclic group as its objects. Translates of C give edges from $a^k H_u$ to $a^k H_v$. Translates of D_v give edges from $a^k H_u$ to $a^{k-1} H_v$. Here $a \neq 1$, and distinct powers

represent distinct cosets until the order of a is reached, by (4). These two matchings form a cycle of length $2|a|$ if a has finite order, and a doubly infinite path otherwise. Their union is connected and spans the residue. Possible additional edges cannot destroy connectedness. This also explains the order-three diagram of source Example 4.10; reversing both cyclic coordinates changes the sign of the shift without changing the argument.

Finally every flag of proper type can be normalized to F_J, so the conclusion holds for all residues, including the empty flag's residue. □

Corollary 5.2. *Every rank-three system from a nondegenerate C^+ marking is a firm, residually connected geometry.*

Proof. Every incident pair is a translate of a base pair and extends to a chamber. Objects and the empty flag extend too; any flag of size three is already a chamber. Thus it is a geometry. Every proper flag has rank at most two, so the coset action is transitive on its type. Theorem 5.1 applies. □

This is source Proposition 4.11. It requires neither chirality nor a full automorphism computation.

For rank two, the nontrivial cyclic marking has both parabolics trivial and gives disjoint edges. The full written proof and literal check in [5] refute the unrestricted Lemma 4.8 and Corollary 4.9. Imposing rank at least four in a proposed replacement for Corollary 4.9 produces a different assertion, unproved here. A false universal statement is not recovered by silently changing its hypotheses, and its counterexample does not automatically create a new source obligation for every such replacement.

6 Why the two acting groups differ

The following retained argument pinpoints the principal error. It adds a structural explanation to the explicit counterexample.

Theorem 6.1. *If the marked coset system is a thin geometry, then $\Omega = GC \dot\cup GD_0$ is one gallery component. It contains every object and every incident pair. The stabilizer of C in the full type-preserving group A is trivial. If A is chiral, then $\mathrm{Stab}_A(\Omega) = G$ and the gallery components are indexed by A/G.*

Proof. Thinness makes the i-neighbour of gC equal to gD_i, which lies in GD_0. The i-neighbour of gD_0 is $g\alpha_i C$. Thus Ω is closed under adjacency. The two-step galleries $C, D_0, \alpha_i C$, their reversals, and their translates join C to every gC, because the α_i generate G. Each gD_0 is adjacent to gC. Consequently Ω is a connected component.

Every object occurs in some gC. For an incident pair, an element of the intersection of the two cosets is a common representative, placing the pair in a translate of C. An automorphism fixing C fixes its neighbours, since each panel has a unique other completion. Repetition along galleries fixes Ω pointwise. Since Ω contains every object, the automorphism is the identity. The coset action itself is faithful because its kernel fixes C and lies in $\bigcap_i H_i = P_\varnothing = 1$.

Suppose now A is chiral. If $f \in A$ preserves Ω, then fC lies in GC or GD_0. The second possibility would identify the two A-orbits of the adjacent chambers C, D_0. Hence $fC = gC$ for some $g \in G$. The element $g^{-1}f$ fixes C and is the identity, so $f = g$. Conversely G preserves Ω. Every chamber is an A-translate of C or D_0, so every component is an A-translate of Ω. The orbit-stabilizer bijection gives A/G. □

The theorem does not assert that A/G has one element. The accepted example [4] has $|G| = 1092$, $|A| = 2184$, two gallery components, and a disconnected rank-two residue. Its full-group chirality supplies the hypotheses of the published Theorem 1.1 and disproves its

conclusion. The switch from proper-flag transitivity for A to that property for G is therefore a substantive error, separate from the smaller missing rank and firmness conditions.

7 Disposition of the source

Figure 1 illustrates the smaller counterexample from [6]. Its chambers are the four triangles $C_j = \{c_j, x_j, y_j\}$ and $D_j = \{d, x_j, y_j\}$, for $j = 1, 2$. The full typed group exchanges the two halves. Its chamber orbits are $\{C_1, C_2\}$ and $\{D_1, D_2\}$, and the only adjacencies join C_j to D_j. Yet the panel $\{c_j, x_j\}$ has only one completion. This is the missing-firmness issue; it is separate from the principal $\mathrm{PSL}_2(13)$ counterexample.

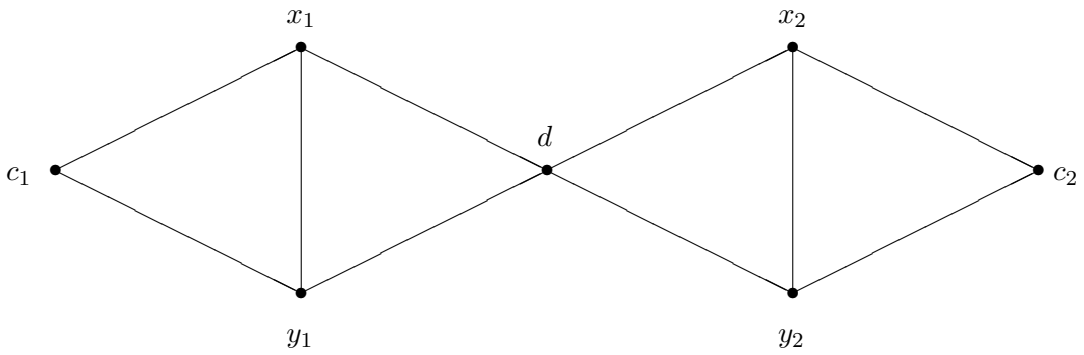


Figure 1: All incidences of the seven-object example. The type classes are $\{d, c_1, c_2\}$, $\{x_1, x_2\}$ and $\{y_1, y_2\}$.

The complete source ledger accompanies this manuscript. The following table records the mathematical outcomes; it is not a claim that false statements have been reproved.

Source assertion	**Disposition and proof**
Theorem 1.1	Refuted in its literal full-group scope by the accepted five-page counterexample [4]. The valid specified-action criterion is Theorem 5.1.
Proposition 2.1	Cited Tits construction; object and pair normalization proved in Section 2.
Theorem 2.2	Cited flag-transitive residue theorem from the source's reference [1]. Its sufficient part needed here is proved in Section 3; the whole cited monograph is not an additional intake.
Theorems 3.1, 3.3; Corollary 3.4	Explicitly borrowed C-group and C^+ correspondence results from the source's reference [6], *Highly symmetric hypertopes*. Their attribution is retained; the present corrections do not refute them.
Construction 3.2	The parabolics are identified with $P_{I\setminus\{i\}}$ in Section 4; left and right cosets are related by the explicit inversion map in Section 1.
Chirality implies thinness; Theorem 4.1; Proposition 4.2	False without firmness under the stated geometry definition; the seven-object example [6] supplies the full group and all residues explicitly. The firm normalization statements are proved in Proposition 4.2.
Lemma 4.3; Remark 4.4	The cited internal map, its object-surjectivity criterion, and its distinction from an isomorphism are proved in Section 2 for the interfaces used here.

Lemma 4.5	Recovered as Lemma 2.2, with the smaller flag-type transitivity present.
Proposition 4.6	Refuted by the retained C_2^2 example in Section 2. Adding smaller-type transitivity repairs the equivalence.
Proposition 4.7	The displayed rank-one boundary is false, as shown in [6]; the preceding source paragraph focuses on rank at least two. Proposition 3.2 states the valid rank and action hypotheses.
Lemma 4.8; Corollary 4.9	Refuted in unrestricted rank by [5]. The canonical-neighbour lemma for rank at least three is Lemma 4.1. General firmness in rank at least four without further action hypotheses remains unproved.
Example 4.10; Figure 1	The cyclic two-matching construction is proved for all cyclic orders in Theorem 5.1; the source's order-three illustration is a special case.
Proposition 4.11	Recovered as Corollary 5.2.

No named open question is solved or added by this disposition. The principal refutation, valid source results, missing hypotheses, and useful unproved variants are separate outcomes. Existing computational certificates are reused without repeating their acquisition. The exact finite checks in [5, 6] verify their stated examples, not the arbitrary-group proofs above. Compression measurements and further Lean formalization are deferred.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162.

[2] Human Compression project, *Incidence results derived from relation matrices*, retained proof bank, Section 1.

[3] Human Compression project, *Chamber components in marked coset geometries*, retained phase276 manuscript, Sections 1–5.

[4] Human Compression project, *A chiral coset geometry with disconnected residues*, accepted five-page manuscript and exact certificate.

[5] Human Compression project, *Panels in the coset construction for a C^+-group*, accepted three-page manuscript, with the rank boundary and the retained finite panel checks.

[6] Human Compression project, *Chirality without firmness*, accepted two-page manuscript and seven-object certificate.